SECTION A

MULTIPLE CHOICE QUESTIONS (20 MARKS)

(INSTRUCTION: Please answer all 20 questions in the answer sheet on page 13)

1) Which statement is referring to the forwarding function in the network layer?
   A) Packet forwarding does not require input buffering.
   B) The transfer of packet from an incoming link to an outgoing link within a single router.
   C) Determine the paths that packets take on their trips from source to destination.
   D) The RIP, OSPF and IS-IS are the protocols used to transfer the packets.

2) Computer networks that provide only a connectionless service at the network layer are called _____.
   A) Datagram Networks
   B) Virtual-Circuit Networks
   C) ATM
   D) Frame relay

3) Which of the following is **NOT** a switching technique in switch fabric for the input port processing in the router?
   A) Memory.
   B) Bus.
   C) Crossbar.
   D) Star.

4) As far as the outside world is concerned, all the followings are **FALSE** about Network Address Translation (NAT), **EXCEPT**:

A) More than one IP address is needed from the ISP for all devices in a network.

B) Local administrator can't change addresses of devices in his network without notifying outside world.

C) Local administrator can change ISP without changing addresses of devices in local network.

D) All local addresses are visible by outside world.

5) All the following statements are **TRUE** about Internet Control Message Protocol (ICMP), **EXCEPT**:

A) ICMP is used by ping for echo request/reply.

B) ICMP is used for Error correction.

C) ICMP uses UDP at transport layer.

D) ICMP is used for Error reporting.

6) To transmit a video to a set of 10 different hosts on a large network, what type of routing would be best fit?

A) Multicast.

B) Broadcast.

C) Unicast.

D) Anycast.

7) What is the reason link state based protocols are not used for inter-domain routing?

A) They do not scale.

B) Networks share topology information.

C) They are hierarchical structured.

D) They allow autonomous administration.

8) Protocol Data Unit (PDU) for Layer 2 is known as _____.

A) message

B) segment

C) datagram

D) frame

9) These are services offered by link layer protocol, **EXCEPT**:

A) Error detection.

B) Error correction.

C) Connection establishment.

D) Flow control.

10) In multiple access protocols, computer networks regulate their transmission through shared **X** channel. What is **X**?

A) Anycast.

B) Multicast.

C) Unicast.

D) Broadcast.

11) Which MAC protocol is used in wireless LAN?

A) Slotted ALOHA.

B) Unslotted ALOHA.

C) CSMA/CA.

D) CSMA/CD.

12) How big is the MAC address space?

A) 2^{16}

B) 2^{32}

C) 2^{48}

D) 2^{128}

13) Which of these is **NOT** the content of ARP table?

A) IP address.

B) MAC address.

C) TTL value.

D) RTT value.

14) A switch is _____.

A) a layer 2 device

B) a layer 3 device

C) an application layer device

D) a physical layer device

15) Which of these is **NOT** a feature of a switch?

A) It stores and forwards Ethernet frames.

B) A switch must be configured for it to work.

C) A switch can handle a mixture of speeds from different hosts.

D) A switch can send frames to all its directly connected hosts without collision.

16) Which of these is the more economical and scalable way to connect 3 Virtual LANs spanning 2 switches?

A) Use 2 cables to connect each switch.

B) Use 3 cables to connect each VLAN in the different switches.

C) Use VLAN trunking between the different switches.

D) Use a router attached between the 2 switches.

17) Which of the following characteristic is **FALSE** about ad hoc wireless network?

A) Requires at least a base station.

B) Node can only transmit to other nodes within link coverage.

C) Nodes organize themselves into a network.

D) Wireless nodes have no such infrastructure with which to connect.

18) Which of these examples is considered as *multiple hops* and *infrastructure-based* wireless network?

A) Piconet.

B) MANET.

C) WiFi.

D) Bluetooth.

19) Which of the following are components that can be found in both GSM 2G and 3G cellular network architecture?

A) Base Transceiver Station (BTS) and Radio Network Controller (RNC).

B) Base Station Controller (BSC) and Gateway Mobile Switching Center (MSC).

C) Gateway GPRS Support Node (GGSN) and Mobile Switching Center (MSC).

D) Mobile subscribers and Radio Network Controller (RNC).

20) All of the following scenario are describing a spectrum of mobility for a mobile wireless user from network perspective **EXCEPT**:

A) He is playing a *Dota* online game on notebook in his room.

B) He keep connecting or disconnecting his notebook from network using DHCP.

C) He is taking a laptop from his office to a meeting room.

D) He is calling someone via mobile phone while driving.

SECTION B

STRUCTURED QUESTIONS (80 MARKS)

(INSTRUCTION: Please answer ALL questions in the answer booklet provided.)

QUESTION 1 [35 Marks]

(a) The following questions are related to the functions of a router.

i) Describe the switching techniques of bus and crossbar. Which technique minimizes packet contention? [**5 marks**]

ii) Describe how a Head-Of-the-Line (HOL) may occur inside an input queued switch as in Figure 1. Complete the diagram to show the forwarding direction and occurrence of HOL. [**3 marks**]

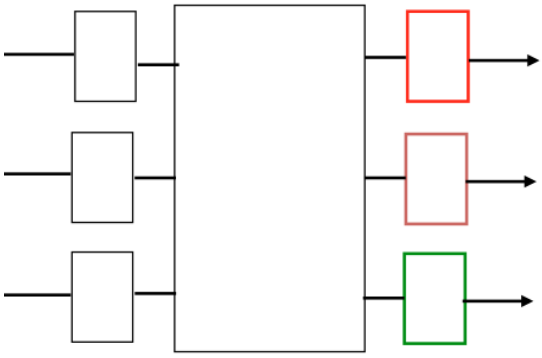


Figure 1: Input and output queued switch

(b) Assume a Maximum Transfer Unit (MTU) for all routers is 1300 bytes. If the incoming datagram size is 4500 bytes. Determine the number of fragments and draw all the fragment blocks according to the following datagram format in Figure 2. [**5 marks**]

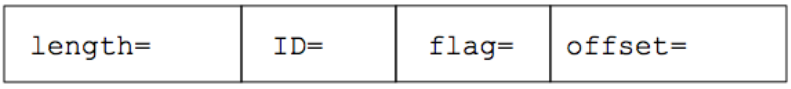


Figure 2: IP fragmentation

(c) Figure 3 shows a network address design using classful and classless (CIDR) addressing scheme for an organization. Answer the following.

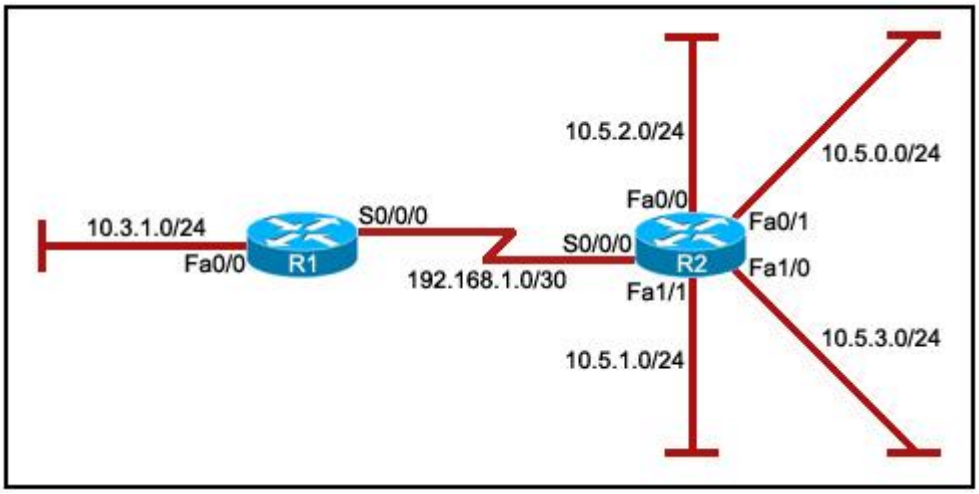


Figure 3: Network address design

i) List THREE (3) subnet addresses that implement classful addressing scheme. What class does this address is categorized? [**2 marks**]

ii) State one subnet address that uses CIDR. What is the subnet mask value? [**2 marks**]

iii) An ISP assigns a host in a subnet the address 10.3.1.100/24. Suppose this subnet only has 200 hosts. Calculate: [**4 marks**]

- the subnet mask (dotted decimal)
- the subnet address
- the number of addresses still unused

(d) Figure 4 shows the four-step process involved in the DHCP client-server interaction when a newly arrived host acquires network setting from DHCP server. Give the four steps in the CORRECT ORDER. [**2 marks**]

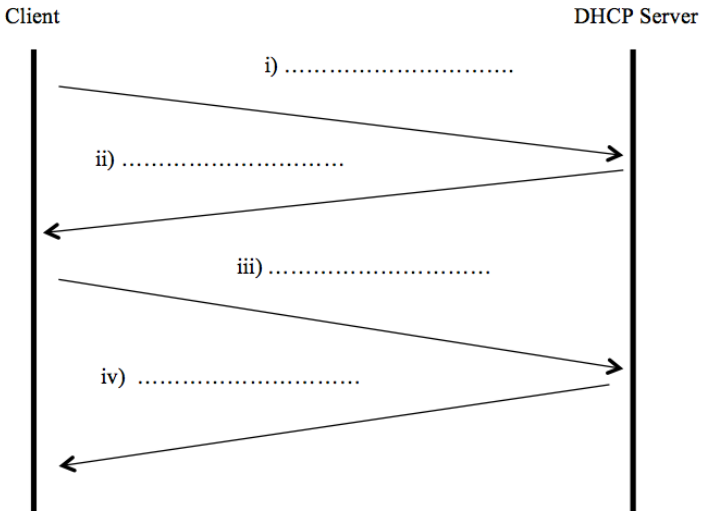


Figure 4: DHCP client-server interaction.

(e) Consider a network topology as in Figure 5. Answer the following questions.

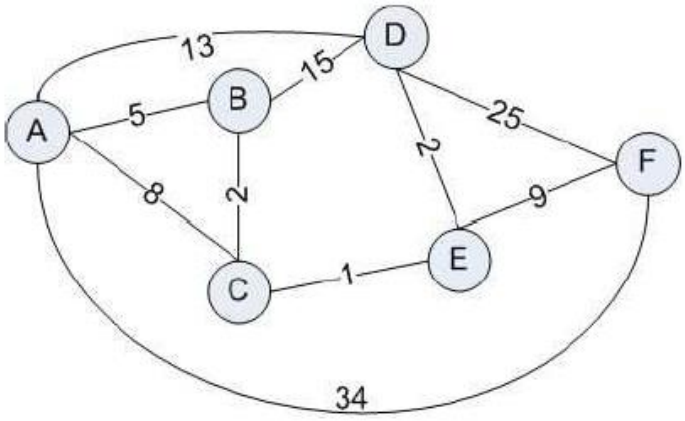


Figure 5: Routers and link cost value

i) Reproduce Table 1 in your answer booklet to show the operation of Dijkstra's algorithm in computing least cost path from A to F. [**7 marks**]

Table 1: Least cost path

Step	N'	*D(B), p(B)*	*D©, p©*	*D(D), p(D)*	*D(E), p(E)*	*D(F), p(F)*
0	*A*	5, A	8, A	13, A	∞	34, A

ii) Assume Table 2 is the result of an operation of Dijkstra's on Figure 6. Construct forwarding table for node E. Determine least cost paths from node E to A. [**5 marks**]

Table 2: Least cost path

N'	*D(A), p(A)*	*D(B), p(B)*	*D©, p©*	*D(D), p(D)*	*D(F), p(F)*
E	∞	10, E	∞	3, E	2, E
EF	∞	5, F	∞	3, E	-
EFD	7, D	5, F	5, D	-	-
EFDC	7, D	5, F	-	-	-
EFDCB	7, D	-	-	-	-
EFDCBA	-	-	-	-	-

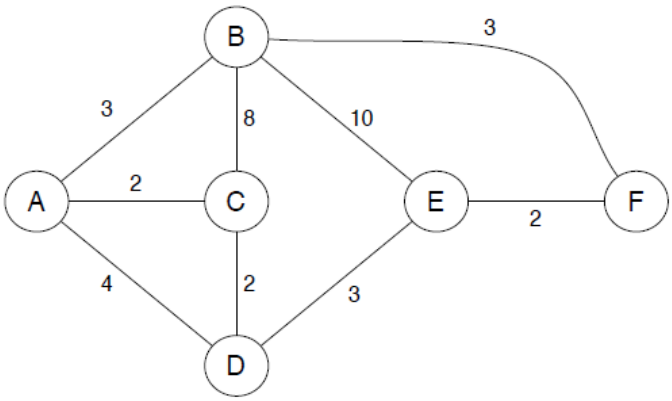


Figure 6: Routers and link cost value

QUESTION 2 [30 Marks]

(a) Given a generator bits = `1010`, and transmitted data is `111010110`. The receiver use cyclic redundancy check to detect the error. Determine whether the transmitted data has error or not? Show your works. [**5 marks**]

(b) State THREE significant characteristics for TDMA and FDMA. [**6 marks**]

(c) Table 3 describes taxonomy of Media Access Control (MAC) taxonomy. MAC protocols can be divided into three categories and each category may use different protocols. Reproduce Table 3 in your answer booklet with the correct answers. List two examples protocols in each MAC category. [**4 marks**]

Table 3

Media Access Control						
Categories						
Protocols						

(d) Assume that all ARP caches have entries of all the MAC addresses as shown in Figure 7. Reproduce Table 4 below in your answer booklet with the correct answer by indicating the source/destination MAC and IP addresses at chosen points during the transmission. [**4 Marks**]

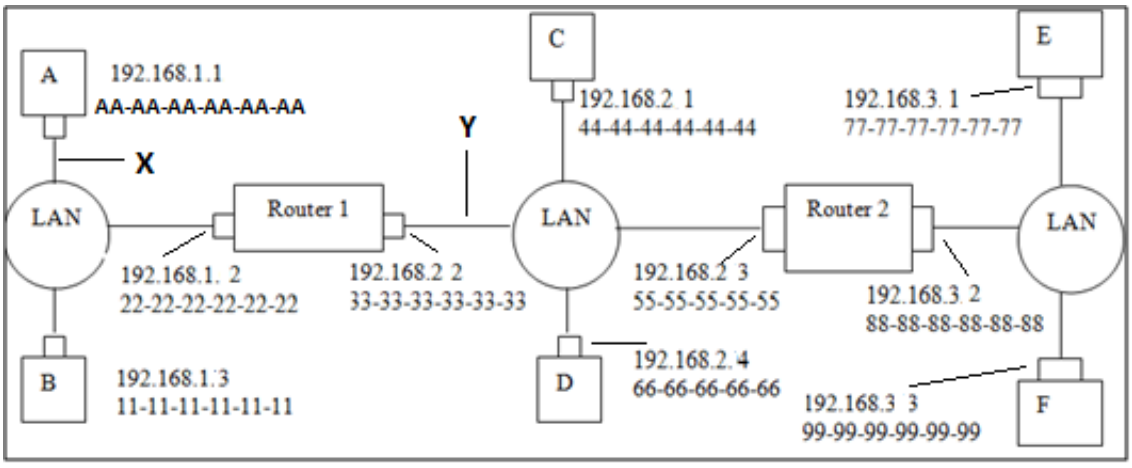


Figure 7

Table 4

Point	Sender	Receiver	Source IP address	Destination IP address	Source MAC address	Destination MAC address
X	PC-A	PC-C				
Y	PC-B	PC-F				

(e) Suppose now A would like to send an IP datagram to B, and all of their ARP caches are empty. Explain what happened in the network until A is able to send the datagram. [**4 Marks**]

(f) The switch in Figure 8 is attached to the different hosts as shown. Host PC1 wants to communicate with PC4. Reproduce Table 5a and Table 5b in your answer booklet with the appropriate steps taken to update switch table and experienced by the switch in this communication. Initially the MAC address table is empty. [**7 Marks**]

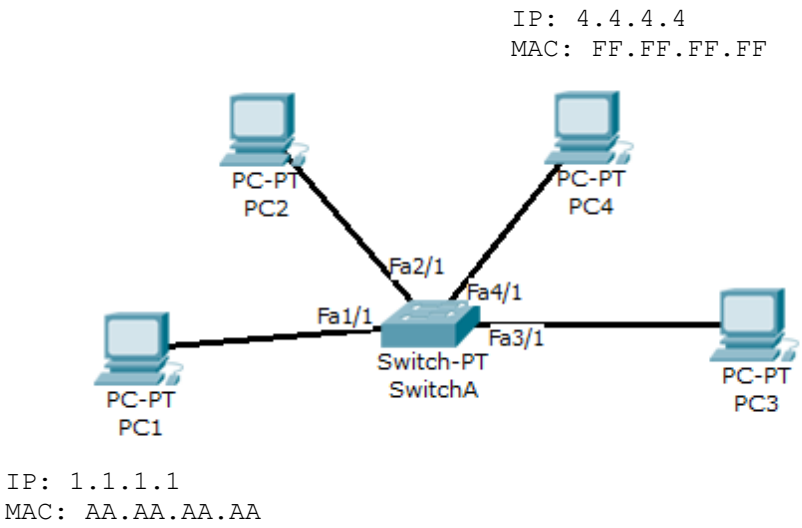


Figure 8

Table 5a

Step #	Step Description

Table 5b

MAC Address	Interface/Port

QUESTION 3 [15 Marks]

(a) Consider the given statements in Figure 9a. Based on the given statements and your knowledge of mobile and wireless network, illustrate or draw the GSM 2G cellular network architecture. You may use the key components in Figure 9b. [**10 marks**]

The region covered by a cellular network is partitioned into a number of geographic coverage areas, known as cells. Each cell contains base transceiver station. Base station controller service several tens of base transceiver stations. The base station controller and its controlled base transceiver stations collectively form a GSM base system station.

Figure 9a

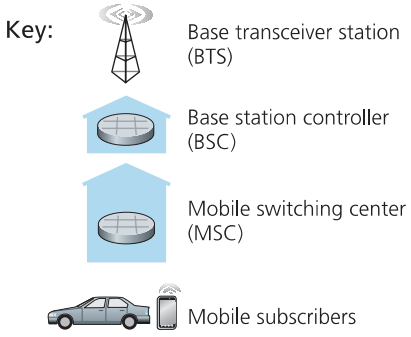


Figure 9b

(b) The following Table 6 shows the definition of terminologies used in wireless network. Answer (i) to (v) by choosing the correct terms from the list below. [**5 marks**]

Visited network	*Foreign agent*	*Permanent address*
No mobility	*High mobility*	*Home agent*
Home network	*Correspondent*	*Care-of-address*

Table 6

Terminolgy	Definition Description
i) ……	Mobile user, passing through multiple access point while maintaining ongoing connections.
ii) ……	Network of cellular provider you subscribe to.
iii) ……	Address in home network, *can always* be used to reach mobile.
iv) ……	Network in which mobile currently resides.
v) ……	Address in visited network.

------- End of Question -------

GOOD LUCK !

ANSWER SHEET FOR SECTION A

NAME :				MARKS
MATRIC NO.:		SECTION :		

Example: =A=  =C= =D=

1)	=A=	=B=	=C=	=D=
2)	=A=	=B=	=C=	=D=
3)	=A=	=B=	=C=	=D=
4)	=A=	=B=	=C=	=D=
5)	=A=	=B=	=C=	=D=
6)	=A=	=B=	=C=	=D=
7)	=A=	=B=	=C=	=D=
8)	=A=	=B=	=C=	=D=
9)	=A=	=B=	=C=	=D=
10)	=A=	=B=	=C=	=D=
11)	=A=	=B=	=C=	=D=
12)	=A=	=B=	=C=	=D=
13)	=A=	=B=	=C=	=D=
14)	=A=	=B=	=C=	=D=
15)	=A=	=B=	=C=	=D=
16)	=A=	=B=	=C=	=D=
17)	=A=	=B=	=C=	=D=
18)	=A=	=B=	=C=	=D=
19)	=A=	=B=	=C=	=D=
20)	=A=	=B=	=C=	=D=